

MNIST NOTEBOOK COMPARISON

VERDICT: `mnist.ipynb` is the better-working notebook. It keeps training and evaluation in the same 0-1 pixel scale and records about 97.6% test accuracy.

A. `mnist.ipynb` | BETTER WORKING

- Two focused code cells: train/evaluate/save, then webcam prediction.
- Uses Keras MNIST and divides both training and test pixels by 255.0.
- Recorded result: about 97.6% test accuracy and 0.0769 loss.
- Webcam input is also normalized by 255.0 before prediction.
- Simpler to rerun and easier to audit because the data pipeline is consistent.

Best use: dependable baseline training and live digit inference.

B. `read-mnist-dataset.ipynb` | BROADER, BUT BROKEN

- Thirteen cells: custom IDX loader, image display, training, webcam, and evaluation.
- Successfully demonstrates reading the repository's raw MNIST files from data/archive.
- Main issue: trains on raw pixels 0-255, then evaluates/predicts on pixels scaled to 0-1.
- Recorded evaluation is near random: about 9.74% accuracy and 2.4351 loss; sample predictions collapse to 8.
- Can overwrite the healthy shared `mnist_model.keras` file when run after the better notebook.

Best use: dataset-format learning, after fixing normalization before `model.fit()`.

*ANALOGY: The model is a scale trained to weigh ingredients in kilograms. **`mnist.ipynb`** uses kilograms every time; **`read-mnist-dataset.ipynb`** trains in grams but tests in kilograms, so the same number looks completely different to the scale.*

CODE ISSUES AND RECOMMENDED FIXES

- Add `x_train = np.array(x_train, dtype='float32') / 255.0` and `x_test = np.array(x_test, dtype='float32') / 255.0` before `model.fit()`.
- Evaluate the model trained in the same notebook instead of relying on a shared `mnist_model.keras` that another notebook may overwrite.
- Use absolute or notebook-relative paths deliberately; webcam cells also require a camera and GUI support.